

Solution Requirements

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being - HDI Predictor
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>No authentication required - open access web application</i>	Low
2	Authorization levels	<i>Single user access - no roles or permissions needed</i>	Low
3	External interfaces	<i>Kaggle HDI dataset used as external data source</i>	High

		<i>services the system must connect to]</i>	
4	Transactions processing	<i>User submits input form, model processes and returns HDI score</i>	High
5	Reporting	<i>Display HDI score and development category on result page</i>	High
6	Business rules, etc.	<i>HDI ≥ 0.8=Very High, 0.7-0.8=High, 0.55-0.7=Medium, <0.55=Low</i>	High
7	Compliance to laws or regulations	<i>Uses open source UNDP HDI dataset from Kaggle</i>	Medium
8	<i>Other</i>	<i>Model accuracy must be above 90% R2 score</i>	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>Prediction response time under 2 seconds</i>	<i>Page loads in < 2 seconds</i>
2	Scalability	<i>Handles all 195 country HDI predictions accurately</i>	<i>Supports 195 countries dataset</i>
3	Security & Data Privacy	<i>No personal data collected or stored in the system</i>	<i>No PII stored, open access</i>
4	Reliability & Availability	<i>Application runs without crashes during normal usage</i>	<i>99% uptime on local server</i>
5	Usability & Accessibility	<i>Simple form-based interface, easy for any user to operate</i>	<i>User-friendly, no training needed</i>
6	Other	<i>ML model R2 score must be above 95% for reliable predictions</i>	<i>R2 Score = 97.73%</i>